Lab Report: Measuring the Acceleration Due to Gravity with a Simple Pendulum

Student Name
Lab Partner: Partner Name

PHYS 101, Section 02
Instructor: Dr. Instructor Name

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Abstract

The period of a simple pendulum was measured for six string lengths between 0.20 m and 1.20 m in order to determine the local acceleration due to gravity, g. A linear fit of T^2 against L gave $g = (9.78 \pm 0.12)\,\mathrm{m\,s^{-2}}$, which agrees with the accepted value of $9.81\,\mathrm{m\,s^{-2}}$ within one standard uncertainty. The main source of error was the timing reaction of the operator.

1 Introduction

The purpose of this experiment was to determine g from the dependence of a pendulum's period on its length. For small angular displacements, the period T of a simple pendulum of length L is given by

$$T = 2\pi\sqrt{\frac{L}{g}} \qquad \Longrightarrow \qquad T^2 = \frac{4\pi^2}{g}\,L, \tag{1}$$

so a plot of T^2 against L should be a straight line through the origin with slope $4\pi^2/g$ [1]. Equation (1) holds to better than 0.5 % for amplitudes below about 10° [2].

2 Materials and Methods

2.1 Materials

- Steel bob (50 g) on a light inextensible string
- Retort stand, clamp and split cork

- Metre rule ($\pm 0.5\,\mathrm{mm}$) and digital stopwatch ($\pm 0.01\,\mathrm{s}$)
- Protractor

2.2 Procedure

1. The string was clamped between the cork halves and the length L from the pivot to the centre of the bob was measured with the metre rule.
2. The bob was displaced by approximately 5° and released from rest.
3. The time for 20 complete oscillations was recorded and divided by 20 to obtain the period T; this was repeated three times and averaged.
4. Steps 1–3 were repeated for six lengths between $0.20\,\mathrm{m}$ and $1.20\,\mathrm{m}$.

3 Results

The averaged measurements are summarised in Table 1.

Table 1: Pendulum length, mean period of 20 oscillations and squared period.

L (m)	T (s)	T^2 (s^2)
0.200	0.898	0.806
0.400	1.270	1.613
0.600	1.556	2.421
0.800	1.797	3.229
1.000	2.009	4.036
1.200	2.201	4.844

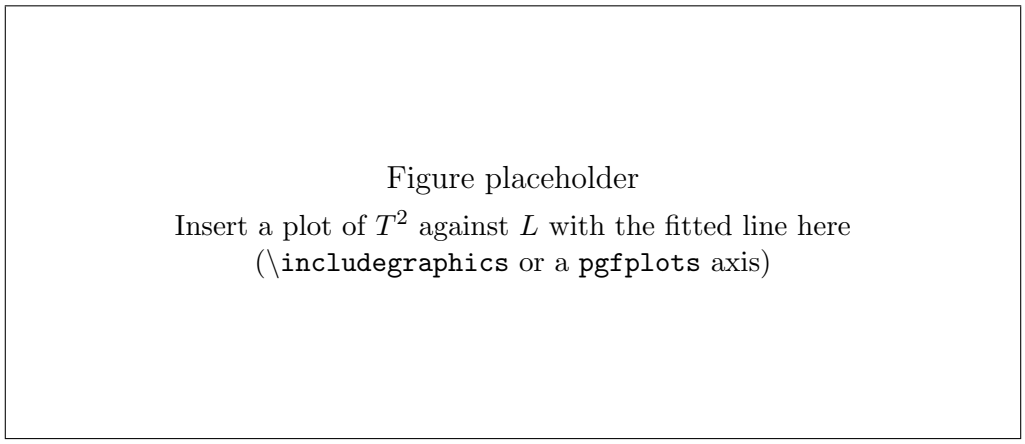


Figure 1: Squared period T^2 as a function of pendulum length L. The solid line is a least-squares fit of Equation (1).

A least-squares fit of T^2 against L (Figure 1), with each point weighted by the timing uncertainty of 0.014 s in T (Appendix A), gave a slope of $(4.038 \pm 0.050)\,\mathrm{s^2\,m^{-1}}$ and an intercept of $(0.00 \pm 0.03)\,\mathrm{s^2}$. From Equation (1),

$$g = \frac{4\pi^2}{\text{slope}} = \frac{4\pi^2}{4.038\,\mathrm{s^2\,m^{-1}}} = (9.78 \pm 0.12)\,\mathrm{m\,s^{-2}}. \tag{2}$$

4 Discussion

The measured value of g agrees with the accepted value of $9.81\,\mathrm{m\,s^{-2}}$ to within 0.3 %, and the accepted value lies inside the quoted uncertainty. The dominant source of uncertainty was the operator's reaction time (about 0.2 s at each start and stop), which was reduced but not eliminated by timing 20 oscillations rather than one. The finite size of the bob and the small but non-zero mass of the string both shift the effective length slightly; timing with a photogate and using a thinner string would reduce these effects. The fit intercept of $(0.00 \pm 0.03)\,\mathrm{s^2}$ is consistent with zero, as predicted.

5 Conclusion

Using a simple pendulum and a plot of T^2 against L, the acceleration due to gravity was found to be $g = (9.78 \pm 0.12)\,\mathrm{m\,s^{-2}}$, in agreement with the accepted value. The method confirms the small-angle relation in Equation (1) and could be improved most by replacing the hand-operated stopwatch with an electronic timer.

References

[1] D. Halliday, R. Resnick and J. Walker, *Fundamentals of Physics*, 10th ed. Wiley, 2013, ch. 15.

[2] R. A. Nelson and M. G. Olsson, "The pendulum—rich physics from a simple system," *American Journal of Physics*, vol. 54, no. 2, pp. 112–121, 1986.

A Sample Calculation

For $L = 0.600\,\mathrm{m}$ the three timed runs for 20 oscillations were 31.08 s, 31.15 s and 31.13 s, giving a mean of 31.12 s and $T = 31.12/20 = 1.556\,\mathrm{s}$. The standard deviation of the three runs is 0.04 s, or $\sigma_T = 0.002\,\mathrm{s}$ per period, but the operator's reaction time of about 0.2 s at each of the start and stop, combined in quadrature and divided by 20 oscillations, contributes 0.014 s; this larger value was used as the uncertainty in each T. Hence $T = (1.556 \pm 0.014)\,\mathrm{s}$ and, since $\sigma_{T^2} = 2T\sigma_T$, $T^2 = (2.421 \pm 0.044)\,\mathrm{s^2}$.